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AI-BASED VOICE COMPLAINT MANAGEMENT SYSTEM

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the
requirements for the Course of
DSA0306-NATURAL LANGUAGE PROCESSING

to the award of the degree of
BACHELOR OF TECHNOLOGY IN
ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

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SEPTEMBER 2026

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**AI-BASED VOICE COMPLAINT MANAGEMENT SYSTEM USING NATURAL LANGUAGE PROCESSING**” has been carried out by K. Deeksha Sree (192424097) under the supervision of Dr. T. Kumaragurubaran and is submitted in partial fulfilment of the requirements for the B.Tech. Artificial Intelligence and Data Science programme at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I, K. Deeksha Sree (192424097), of the B.Tech. Artificial Intelligence and Data Science programme at SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “**AI-BASED VOICE COMPLAINT MANAGEMENT SYSTEM USING NATURAL LANGUAGE PROCESSING**” is the result of my own bonafide efforts. The work presented herein is original and has been carried out in accordance with academic integrity and engineering ethics. All sources, software, datasets, computational resources and AI-assisted resources used in the project have been appropriately acknowledged. I further declare that the data, results, analysis and findings presented in this report have not been fabricated, falsified or manipulated. I confirm that all applicable ethical, academic, institutional and professional requirements have been followed throughout the planning, implementation, testing and documentation of the project.

Place: Chennai

Date: 07-09-2026

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INDIVIDUAL CONTRIBUTION STATEMENT

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
K. Deeksha Sree (192424097)	Complete AI-Voice Complaint Management System	Developed voice input, speech-to- text, NLP processing, AI classification, priority assignment, department routing and database workflow.	Tested transcription, classification, confidence, priority, routing, database and dashboard outputs.	Contributed to all chapters, module documentation, results, conclusion and final report.	100%

Individual Contribution Statement

Total Contribution: 100%

ABSTRACT

The AI-Based Voice Complaint Management System Using Natural Language Processing is designed to simplify the process of registering, understanding, classifying, routing and managing complaints. Instead of requiring users to type a complaint manually, the system accepts a voice complaint and converts the speech into text. Natural Language Processing techniques are then applied to clean and process the text, extract useful linguistic features and identify the complaint category.

The proposed system consists of four major modules: Voice Input and Speech-to-Text, NLP Text Processing and Feature Extraction, AI-Based Complaint Classification, and Complaint Routing, Report Generation and Database. The prototype interface demonstrates audio recording, voice-to-text conversion, complaint analysis, category prediction, confidence estimation, priority assignment, smart department routing and administrator-side complaint management.

The system can be applied in public services, electricity and utility services, internet and telecom services, municipal services, banking, education administration and customer support. Its objective is to reduce manual complaint handling, improve routing, organize complaint records and provide faster access to complaint status and reports.

The system uses NLP and TF-IDF-based text analysis to identify complaint categories and prioritize complaints efficiently. It provides a faster, organized, and user-friendly approach to complaint management while reducing manual effort and improving response handling.

Keywords: AI, Voice Complaint Management, Natural Language Processing, Speech-to-Text, TF-IDF, Machine Learning, Classification, Department Routing

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description
AI	Artificial Intelligence
NLP	Natural Language Processing
STT	Speech-to-Text
TF-IDF	Term Frequency–Inverse Document Frequency
ML	Machine Learning
GUI	Graphical User Interface
API	Application Programming Interface
DB	Database
UI	User Interface

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Complaint management is an important activity in government offices, utility services, educational institutions, organizations and customer-support environments. Traditional complaint systems commonly depend on manual forms, emails, telephone calls handled by staff or separate departmental records. These approaches require human effort and may make classification, routing, tracking and reporting difficult.

Voice interfaces and Natural Language Processing provide an opportunity to automate the first stages of complaint management. A user can speak a complaint naturally, after which the system converts the speech into text and analyzes the language. The resulting information can be used to classify the complaint, estimate its priority and route it to the relevant department.

1.2 Need for the Project

- Typing long complaints can be inconvenient for users.
- Manual complaint classification consumes staff time.
- Complaints may be routed to the wrong department.
- Large complaint volumes are difficult to organize manually.
- Priority identification may be inconsistent.
- Manual report preparation requires additional effort.

1.3 Problem Statement

There is a need for an intelligent complaint-management system that can accept voice input, convert speech to text, understand the complaint using NLP, classify it into a suitable category, assign a priority, route it to the appropriate department, and store the complaint for future tracking and reporting.

1.4 Project Aim

The aim of the project is to design and develop an AI-based voice complaint management system that combines speech-to-text conversion, NLP, machine-learning-based complaint classification, intelligent department routing, database management and report generation.

1.5 Project Objectives

1. Capture a user's complaint through a microphone.
2. Convert the recorded speech into text.
3. Preprocess complaint text using NLP techniques.
4. Extract relevant text features for AI classification.

5. Classify complaints into service categories.
6. Estimate complaint priority and classification confidence.
7. Automatically route complaints to the appropriate department.
8. Store complaint details and status in a structured database.
9. Provide an administrator dashboard for monitoring and updating complaints.
10. Generate complaint summaries and reports.

1.6 Scope

The scope of the proposed system covers voice-based complaint registration, speech-to-text conversion, NLP preprocessing, feature extraction, AI-based classification, priority assignment, department routing, complaint storage, status management, administrative monitoring and summary reporting.

Included:

- Voice-based complaint registration.
- Speech-to-text conversion.
- NLP preprocessing and feature extraction.
- AI-based category classification.
- Priority assignment and confidence display.
- Department routing.
- Complaint database and status management.
- Administrative dashboard and summary reporting.

Excluded:

- Fully autonomous resolution of every complaint.
- Large-scale deployment without additional infrastructure.
- Collection of unnecessary personal information.

Assumptions and Boundary Conditions:

- The user provides a reasonably clear voice recording.
- The speech-to-text service or component is available.
- The classification model has been trained using representative complaint data.
- The database is available during complaint submission.

1.7 Expected Engineering Outcome

The expected outcome is a working prototype that transforms a spoken complaint into a structured complaint record and displays the transcribed complaint, predicted category, confidence, priority, assigned department and stored status.

The system is expected to demonstrate the practical integration of speech processing, NLP, machine learning, database management and a user interface in a single complaint-management workflow. Testing should verify the correctness of each stage and identify conditions requiring human verification.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The project considers existing approaches to complaint management, speech recognition, Natural Language Processing, text classification, ticket routing and database-supported service management. The review focuses on the engineering requirements needed to connect these technologies into one practical workflow.

2.2 Review of Existing Work

Manual Complaint Systems: Manual complaint systems use registers, written forms or staff-recorded information to capture complaints. They are simple to implement but time-consuming, dependent on staff availability and difficult to analyze at large scale.

Web / Email Complaint Systems: Web forms and email allow users to submit digital complaints that can be stored electronically. Searching and retrieval are easier, but users must type the complaint and manual reading or classification may still be required.

Voice-Based Complaint Processing: Voice-based systems allow users to describe an issue naturally. Speech-to-text technology converts spoken language into a textual representation that can be processed by NLP techniques, creating a bridge between conversational input and automated complaint analysis.

NLP-Based Complaint Classification: NLP classification systems commonly perform text normalization, tokenization, removal of unnecessary words, feature extraction and supervised classification. Features such as TF-IDF vectors can represent important words and phrases numerically, allowing machine-learning algorithms to distinguish complaint categories.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Manual Complaint	Simple, low cost	Time-consuming; manual classification	Baseline
Web / Email	Digital records	Requires typing and manual review	Alternative
Voice + AI	Natural interaction; automated analysis	Depends on speech/model accuracy	Proposed system
Call Centre	Human interpretation	High staff effort	Alternative

Table 1: Comparative Analysis of Existing Complaint Approaches

2.4 Research / Engineering Gap

Existing complaint approaches either depend heavily on manual interaction or separate the submission, classification and routing activities. The engineering gap is an integrated workflow that accepts voice input, converts speech to text, processes the language, classifies the complaint, assigns priority, routes it and stores it for monitoring.

2.5 Proposed Contribution

The proposed system integrates voice input, speech-to-text, NLP analysis, AI classification, priority estimation, department routing, database storage and administrative reporting into a single pipeline. The main contribution is a voice-to-action complaint workflow that converts an unstructured spoken problem into a structured service record.

CHAPTER 3

ENGINEERING DESIGN, TRADE-OFFS, ETHICS, SAFETY AND SECURITY

3.1 Requirements

Stakeholder	Requirement
User	Submit complaint using voice
Administrator	Monitor and update complaints
Department	Receive correctly routed complaints
Management	View summaries and reports

Table 2: Stakeholder Requirements

Requirement	Type	Measurable Target
Record voice complaint	Functional	Audio captured successfully
Convert speech to text	Functional	Readable complaint transcription
Edit transcribed complaint	Functional	Updated text accepted
Preprocess complaint text	Functional	Text processed successfully
Extract NLP features	Functional	Numerical representation generated
Classify complaint category	Functional	Service category predicted
Assign priority/confidence	Functional	Priority and confidence displayed
Route complaint to department	Functional	Validated category-to-department mapping
Store complaint in database	Functional	Record stored successfully
Update complaint status	Functional	Status changes saved
Generate summaries/reports	Functional	Summary information available
Accuracy	Non-Functional	Reliable classification under tested conditions
Reliability	Non-Functional	Consistent processing and storage
Usability	Non-Functional	Simple voice-first workflow
Security and privacy	Non-Functional	Authorized access and minimum necessary data

Table 3: Functional and Non-Functional Requirements

3.2 Design Specification

The system is designed as a modular pipeline. The microphone captures the complaint, the speech-to-text component generates a transcript, NLP processing cleans and normalizes the text, feature extraction converts the text into a numerical representation, and the AI classifier predicts the complaint category. The result is then used for priority estimation, department routing and database storage.

3.3 System Architecture

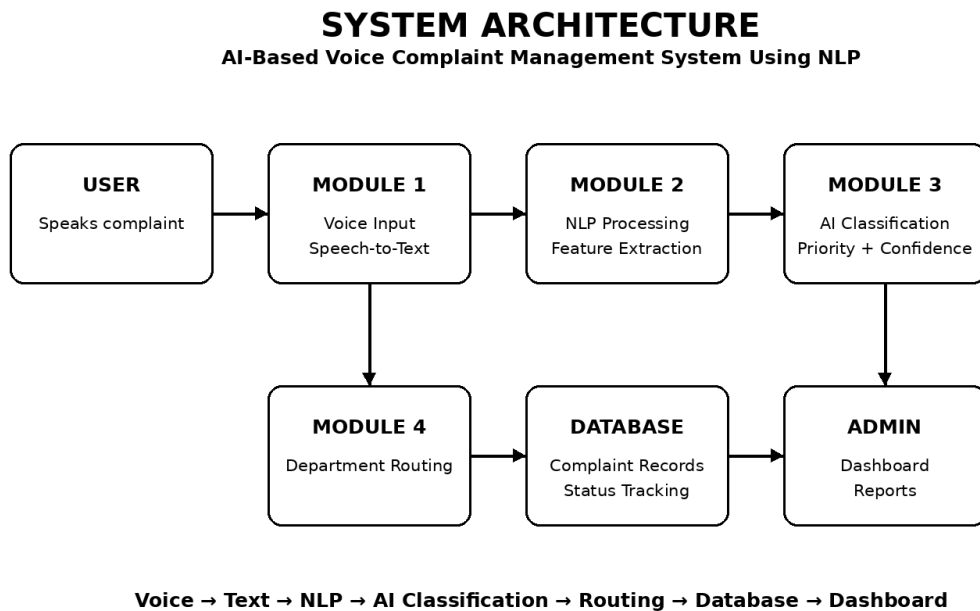


Fig. 1: System Architecture

System Workflow: Voice Input → Speech-to-Text → NLP Processing → Feature Extraction → AI Classification → Priority Estimation → Department Routing → Database → Dashboard / Reports.

3.4 Module Description

Module 1 – Voice Input and Speech-to-Text: The microphone captures the complaint as an audio signal. A speech-recognition component converts the audio into text, and the transcribed complaint is displayed so that the user can review or edit it before analysis.

Module 2 – NLP Text Processing and Feature Extraction: The transcribed text is normalized, tokenized and cleaned. Feature extraction such as TF-IDF converts the processed complaint into numerical features for the classifier.

Module 3 – AI-Based Complaint Classification: A trained machine-learning classifier predicts the complaint category from the extracted text features and returns a confidence value when supported. Priority can be assigned using the project's decision rules.

Module 4 – Complaint Routing, Report Generation and Database: The predicted category is mapped to the responsible department. Complaint details, category, priority, confidence, department, date/time and status are stored in a database and displayed through the administrator dashboard.

3.5 Alternative Solutions and Evaluation

Criterion	Alternative 1 (Manual)	Alternative 2 (Web/Email)	Alternative 3 (Voice + AI)
Automation	Low	Medium	High
Typing effort	High	High	Low
Natural interaction	Low	Medium	High
Automated classification	No	Optional	Yes
Routing support	Manual	Manual/Rule-based	AI + Rule-based
Scalability	Low	Medium	High

Table 4: Alternative Solutions Evaluation Criteria

Voice input is selected because it reduces typing effort and supports natural complaint description. NLP-based classification is selected to automate category prediction. A structured database is selected for searchable records, status management and reporting.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Voice Input	Typed-only input	Natural interaction	Noise/accent sensitivity	Voice input selected to reduce typing effort.
NLP Classification	Manual classification	Fast categorization	Depends on training data	AI classification selected for automation.
TF-IDF Features	Raw text only	Simple and efficient	Limited deep semantics	Suitable baseline feature representation.
Automated Routing	Manual routing	Reduces staff effort	Wrong prediction can misroute	Use validated category-to-department rules.

Table 5: Trade-off Analysis

3.7 Sustainability, Ethics, Safety and Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Digital records reduce paper	Paper use vs digital storage	Selected database-backed workflow.
Ethics	Voice data requires purpose limitation	Privacy and access requirements	Collect only necessary information.
Safety	Recognition uncertainty can affect routing	Confidence and human review	Allow review of low-confidence cases.

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Security	Complaint records require protection	Authentication, access control, backups	Restrict access and protect stored data.

Table 6: Sustainability, Ethics, Safety and Security Considerations

Speech recognition should be tested under realistic noise conditions. Low-confidence or sensitive cases can be flagged for human verification. The system should not claim certainty when the model confidence is low.

3.8 Risk Register

Risk	Likelihood	Severity	Mitigation
Background noise	Medium	Medium	Clear recording conditions and preprocessing
Speech recognition error	Medium	Medium	Allow transcript editing
Incorrect category	Medium	High	Improve training data and confidence thresholds
Incorrect department routing	Medium	High	Validate category-to-department mapping
Duplicate complaint	Medium	Medium	Complaint identifiers and database checks
Data loss	Low	High	Regular database backups
Unauthorized access	Low	High	Authentication and access control

Table 7: Risk Register

CHAPTER 4

IMPLEMENTATION, TESTING AND RESULTS

4.1 Development Approach and Resources

The system is developed using a modular approach. Development begins with voice recording and speech-to-text conversion, followed by NLP preprocessing and feature extraction. The classification model predicts the complaint category. The routing and database module stores the result and makes it available through the administrator dashboard.

Resource	Purpose
Computer / Laptop	System development and testing
Microphone	Voice complaint capture
Python environment	Application development
Speech-to-text component	Voice transcription
NLP libraries	Text processing and feature extraction
Database	Complaint storage and status management

Table 8: Resources Used

4.2 Software Implementation and Modern Tools Used

Tool / Software / Equipment	Purpose	Stage Used
Python	Application development	All stages
Speech Recognition / STT	Voice transcription	Module 1
NLP library	Preprocessing and features	Module 2
Scikit-learn / ML library	Classification	Module 3
Database system	Storage and status	Module 4
Web / GUI framework	User interface and dashboard	Module 4

Table 9: Tools and Software Used

4.3 Implementation Process

1. Open the complaint recording interface.
2. Record the user's complaint using the microphone.
3. Convert the audio into text.

4. Display and optionally edit the transcribed complaint.
5. Preprocess the complaint using NLP.
6. Extract text features.
7. Run the trained AI classification model.
8. Display category, confidence and priority.
9. Map the category to the appropriate department.
10. Store the complaint and status in the database.
11. Display the record on the administrator dashboard.
12. Generate complaint summaries and reports.

4.4 Test Plan and Verification

Test Case	Input	Expected Result	Actual Result	Status
Voice input	Microphone recording	Audio captured successfully	Audio recorded successfully	Pass
Speech-to-text	Spoken complaint	Complaint converted to text	Voice converted successfully	Pass
Transcript editing	Edit text and submit	Updated text accepted	Updated text accepted	Pass
NLP processing	Submit complaint text	Text processed successfully	Text processed successfully	Pass
Classification	Known category test	Category predicted	Electricity	Pass
Confidence	Classification test	Confidence displayed	65.0%	Pass
Priority	Complaint analysis test	Priority displayed	Medium	Pass
Routing	Electricity category	Electrical Department	Electrical Department	Pass

Table 10: Test Plan and Verification

4.5 Results

The prototype outputs demonstrate the complete flow from voice complaint recording to AI analysis and administrative management. In the tested case, audio is recorded successfully and converted to the complaint text “light is not working.” The system identifies Electricity as the category, displays 65.0% confidence, assigns Medium priority and routes the complaint to the Electrical Department. The administrator dashboard displays complaint records with category, department, priority, confidence and status.

Metric / Test Result	Observed Result
Voice recording	Successfully completed
Speech-to-text	Successfully completed
Complaint text	“light is not working”
Predicted category	Electricity
Confidence	65.0%
Priority	Medium
Assigned department	Electrical Department

Table 11: Sample Test Results

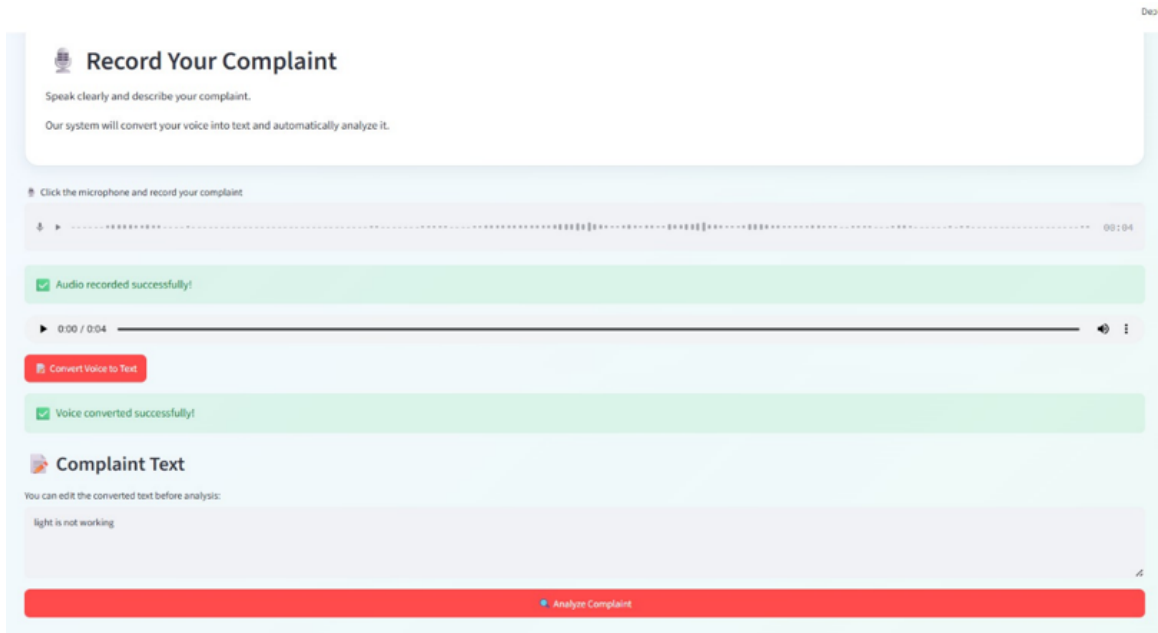


Fig. 2: Voice Complaint Recording and Speech-to-Text Output

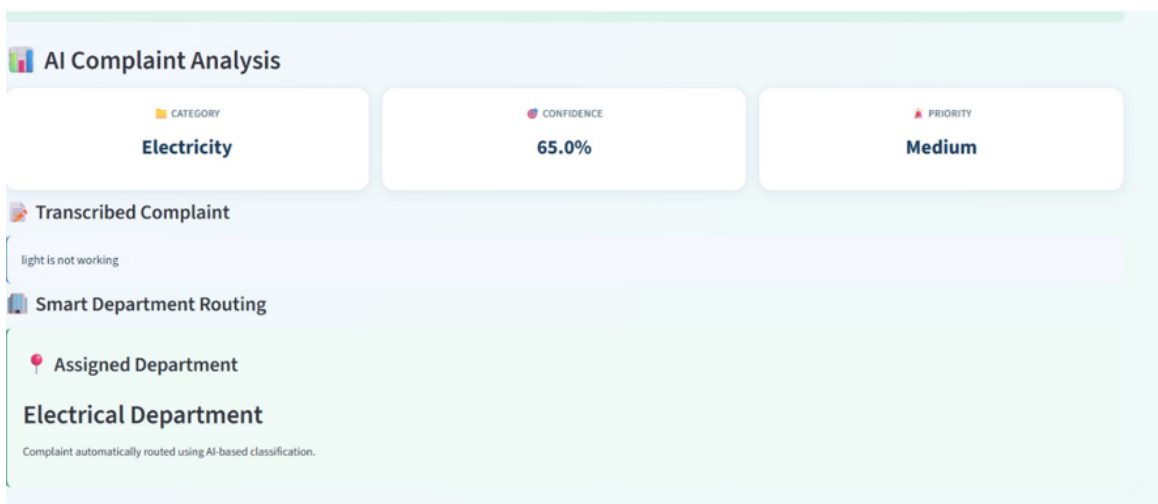


Fig. 3: AI Complaint Analysis and Smart Department Routing

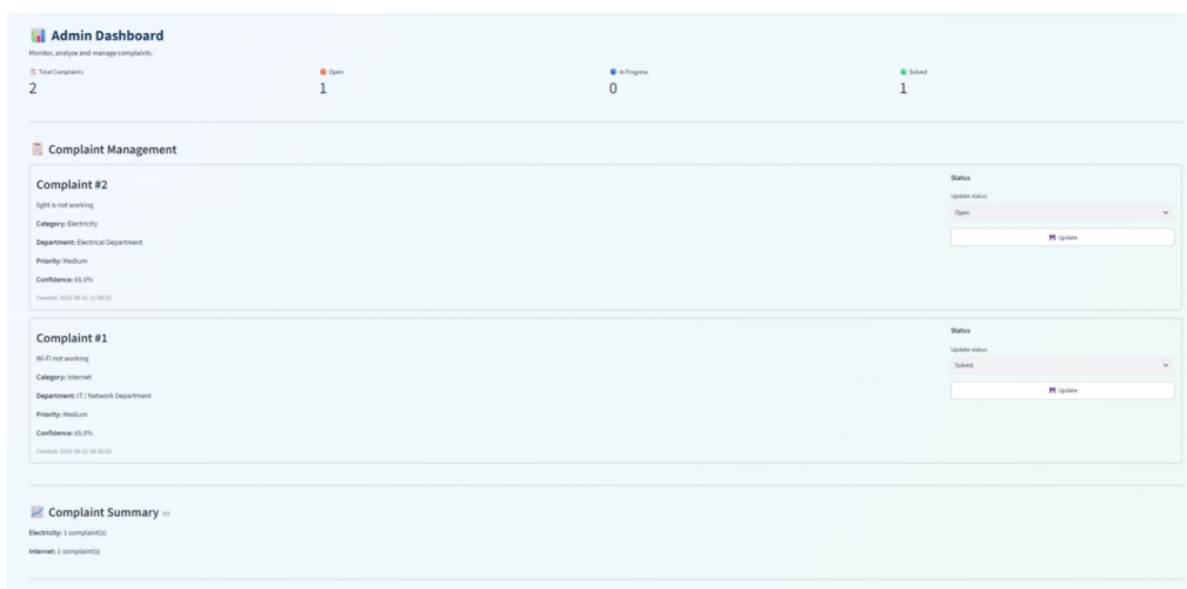


Fig. 4: Administrator Dashboard and Complaint Management

4.6 Data Analysis and Engineering Judgment

System performance is influenced by speech clarity, background noise, training-data quality, text representation, classification accuracy and category-to-department mapping. Allowing the user to review or edit the transcribed complaint provides an additional control before AI analysis. The observed 65.0% confidence is a test-case output and should not be interpreted as the overall model accuracy.

4.7 Achievement of Objectives

Objective	Evidence	Achievement
Capture voice complaint	Microphone recording	Fully Achieved
Convert speech to text	Successful transcription	Fully Achieved
Process complaint text	NLP processing stage	Fully Achieved
Classify complaints	Electricity category output	Fully Achieved
Assign priority/confidence	Medium / 65.0% output	Fully Achieved
Route complaints	Electrical Department output	Fully Achieved
Store/manage records	Administrator dashboard	Fully Achieved
Generate reports	Dashboard / summary output	Partially Achieved

Table 12: Achievement of Objectives

The proposed system improves on manual complaint workflows by reducing typing, automating classification and routing, and providing structured status monitoring. Human review remains important for uncertain predictions and exceptional cases.

4.8 Strengths and Limitations

Strengths:

- Voice-based and user-friendly registration.
- Automatic speech-to-text conversion.
- NLP-based complaint understanding.
- Automated category and priority prediction.
- Smart department routing.
- Centralized complaint records.
- Administrator dashboard and status management.

Limitations:

- Speech recognition can be affected by noise or accents.
- Classification depends on training-data quality and coverage.
- Low-confidence predictions may require human verification.
- Incorrect classification can cause incorrect routing.
- Voice and complaint data require privacy and security controls.

4.9 Project Planning and Management

Phase	Activity
Phase 1	Problem identification
Phase 2	Literature review
Phase 3	Requirements and system design
Phase 4	Voice and speech-to-text module
Phase 5	NLP and feature extraction
Phase 6	AI classification and routing
Phase 7	Database and dashboard integration
Phase 8	Testing and debugging
Phase 9	Results and documentation
Phase 10	Final review and submission

Table 13: Project Milestones

As this Capstone Project was carried out individually, I was responsible for the requirement analysis, design, implementation, testing and documentation of the system under the supervision of Dr. T. Kumaragurubaran.

CHAPTER 5

CONCLUSION AND REFLECTION

5.1 Conclusion

The AI-Based Voice Complaint Management System Using Natural Language Processing provides an automated workflow for collecting and managing complaints. The system combines voice input, speech-to-text conversion, NLP processing, feature extraction, AI-based classification, priority assignment, department routing, database management and an administrator dashboard.

The supplied prototype outputs demonstrate that a spoken complaint can be converted into text, analyzed as Electricity with a displayed confidence of 65.0%, assigned Medium priority, routed to the Electrical Department and managed through an administrative dashboard. This integrated workflow can reduce manual complaint handling and improve organization of complaint records.

5.2 Future Work

- Improve speech recognition for different accents and noisy environments.
- Support multiple Indian and regional languages.
- Use transformer-based NLP models for deeper semantic understanding.
- Improve classification with a larger and more diverse complaint dataset.
- Add automatic duplicate and similar-complaint detection.
- Add SMS, email or app notifications for complaint status updates.
- Provide complaint tracking for users.
- Add advanced analytics and department performance dashboards.
- Deploy the system on cloud infrastructure for large-scale use.
- Strengthen privacy, authentication, audit logs and data protection.

5.3 Professional Learning and Individual Reflection

New Knowledge Acquired:

- Learned practical applications of speech recognition and Natural Language Processing.
- Gained experience with text preprocessing and feature extraction.
- Learned machine-learning-based complaint classification.
- Learned database-supported complaint management.
- Understood the integration of AI modules into a complete application.

Engineering and Professional Skills Developed:

- Python programming.
- Natural Language Processing.

- Machine learning.
- Database management.
- System design and integration.
- Testing and debugging.
- Technical documentation.

Individual Reflection:

During the project, the major engineering challenge was integrating speech-to-text, NLP, classification, routing and database components into a single reliable workflow. The project required attention to input quality, feature representation, model confidence and human verification.

A key engineering decision was to allow the user to review the transcribed complaint before analysis. This provides a practical control against speech-recognition errors. The project provided experience in developing a complete AI-enabled application from problem identification through implementation and testing.

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- [8] Documentation for the database technology used in the final implementation.

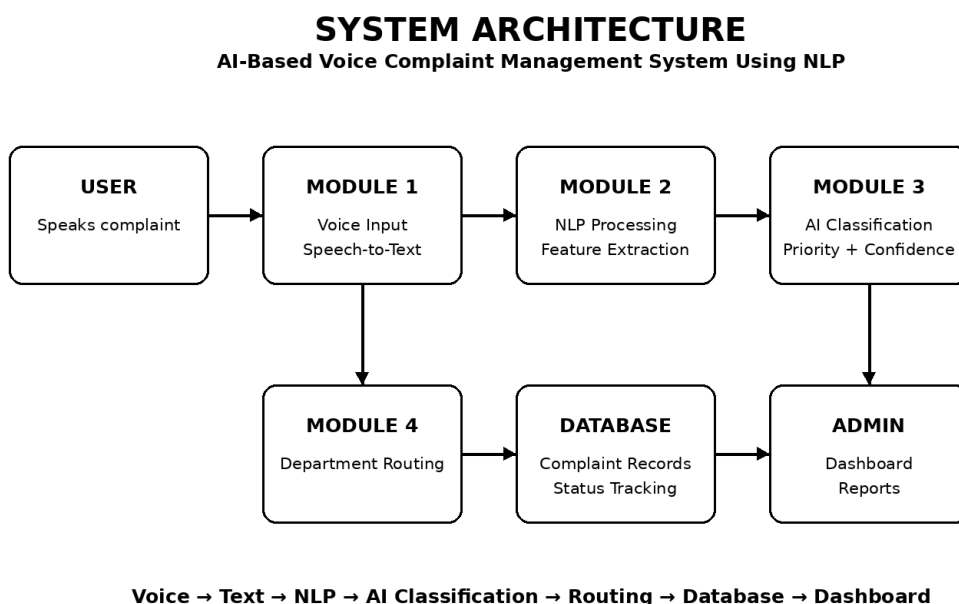
Note: The exact speech-to-text engine, classifier, database technology and software versions should be recorded from the final implementation rather than inferred.

APPENDICES

Appendix A – Detailed System Workflow

Voice Input → Speech-to-Text → NLP → Feature Extraction → AI Classification → Priority Estimation → Routing → Database → Dashboard.

Appendix B – Engineering Drawings / System Architecture



Appendix Figure: System Architecture

Appendix C – Source Code / Code Repository Information

Only essential code excerpts should normally be included in the report. Complete source code may be maintained separately in the approved repository.

Appendix D – Model / Software Information

The exact speech-recognition engine, NLP libraries and classifier used in the final implementation should be recorded with their versions.

Appendix E – Experimental / Raw Data

Test complaints and observed outputs may be maintained as project test data. Personal information should not be included unnecessarily.

Appendix F – Ethics / Institutional Approval

Where applicable, voice and complaint data should be collected and handled according to institutional requirements.

Appendix G – Project Meeting / Progress Records

Project progress records may include requirement discussions, development milestones and testing decisions.

Appendix H – User / Industry Feedback

Where applicable, feedback can be used to improve usability and routing accuracy.

Appendix I – Publications / Patents / Competitions / Product Outcomes

Where applicable.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome	Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	Problem statement and engineering gap	SO1	Ch. 1, Ch. 2
Engineering design under constraints	Requirements, architecture and alternatives	SO2	Ch. 3
Written/oral technical communication	Technical report and presentation	SO3	Whole report + Viva
Ethics and professional responsibility	Privacy, ethics and security considerations	SO4	Ch. 3
Experimentation, analysis and engineering judgment	Testing and results	SO6	Ch. 4 (4.4–4.7)
Independent acquisition of new knowledge	Professional learning and reflection	SO7	Ch. 5 (5.3)
Standards	Design constraints and responsible data handling	SO2	Ch. 3
Sustainability and societal/environmental impact	Digital records and responsible data use	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	Risk register	SO2/SO4	Ch. 3 (3.8)
Security considerations	Access control and data protection	Relevant SO	Ch. 3 (3.7–3.8)
Integrated/system-level engineering	Four-module system workflow	SO1/SO2	Ch. 3 (3.2–3.4)
Project planning and management	Project milestones and individual responsibility	SO5	Ch. 4 (4.9)
Individual contribution	Contribution statement and implementation	SO1–SO7	Contribution Statement + Ch. 4